

Documentation

Introduction & Design

For this project, our group chose a cone with radius 32" and height 36" (or 2.5' and 3', respectively). The equation that we used was

$$\frac{x^2}{900} + \frac{y^2}{900} = \frac{z^2}{1296},$$

where $x \in [-30, 30]$, $y \in [-30, 30]$, and $z \in [0, 32]$.

Construction Process

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Calculations

Precise Theoretic Value

To calculate the precise theoretic value, we will use an integral, specifically a double integral.

Approximate Theoretic Value

To calculate the approximate theoretic value, we will use a Riemann summation, specifically two summations.

Discussion

Precise Theoretic Value

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Approximate Theoretic Value

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